Vehicle Parking Management System

Unit Test Report

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Executed 8 November 2026 on commit 39e0e56. 748 of 748 Jest tests passed in 39 suites.

1. Purpose and scope

This report records the execution of the automated unit and integration tests described in the Test Plan (Sections 3 to 5) against the Sprint 2 and Sprint 3 code. Unit tests (tests/unit) check single modules with no database: validation, fee calculation, password hashing and role checks. Integration tests (tests/integration) call the real Express API through supertest against a MySQL test database, so each request runs through routing, authentication, the service and the repository.

2. Test environment

- Windows 11, Node.js 24.22.0, MySQL 8 (parking_test database, rebuilt from database/schema.sql before the run), Jest 30 with Istanbul coverage, Playwright 1.64.
- Command: npm run test:coverage (jest --runInBand --coverage). Tests run one file at a time because every file resets the shared parking_test database.
- The same suite runs on every pull request and push to main in GitHub Actions; latest: GitHub Actions run 41589654563 on main (3 Nov 2026, after #104 merged), 743 of 743 passed (the 5 tests added in this report's commits came later).

3. Summary

Level	Suites	Tests	Passed	Failed	Time (s)
Unit	13	189	189	0	1.2
Integration	26	559	559	0	229.1
Total	**39**	**748**	**748**	**0**	**230.3**

Result: all 748 tests passed; none failed, skipped or were marked todo.

4. Results by test suite

Test suite	Level	Tests	Passed	Failed	Time (s)
tests/unit/admin.validation.test.js	Unit	11	11	0	0.0
tests/unit/fees.test.js	Unit	12	12	0	0.0
tests/unit/history.validation.test.js	Unit	13	13	0	0.0
tests/unit/parking.validation.test.js	Unit	12	12	0	0.0
tests/unit/password.test.js	Unit	7	7	0	0.7
tests/unit/payments.validation.test.js	Unit	24	24	0	0.0
tests/unit/rates.validation.test.js	Unit	17	17	0	0.0
tests/unit/requireRole.test.js	Unit	5	5	0	0.1
tests/unit/reservations.validation.test.js	Unit	17	17	0	0.0
tests/unit/slots.filter.validation.test.js	Unit	14	14	0	0.0
tests/unit/slots.validation.test.js	Unit	23	23	0	0.0
tests/unit/users.validation.test.js	Unit	13	13	0	0.0
tests/unit/vehicles.validation.test.js	Unit	21	21	0	0.0
tests/integration/admin-users.test.js	Integration	29	29	0	20.1
tests/integration/audit-log.test.js	Integration	10	10	0	6.5
tests/integration/auth.test.js	Integration	12	12	0	6.4

Test suite	Level	Tests	Passed	Failed	Time (s)
tests/integration/authorization.test.js	Integration	15	15	0	1.1
tests/integration/data-protection.test.js	Integration	157	157	0	3.5
tests/integration/exit-awaiting-payment.test.js	Integration	5	5	0	1.3
tests/integration/locate-vehicle.test.js	Integration	18	18	0	16.3
tests/integration/lockout.test.js	Integration	11	11	0	10.7
tests/integration/move-vehicle.test.js	Integration	30	30	0	26.8
tests/integration/parking-history.test.js	Integration	29	29	0	28.3
tests/integration/parking.test.js	Integration	13	13	0	11.8
tests/integration/password-security.test.js	Integration	7	7	0	4.0
tests/integration/payments.test.js	Integration	21	21	0	2.0
tests/integration/rates.test.js	Integration	15	15	0	1.9
tests/integration/reliability.test.js	Integration	18	18	0	3.4
tests/integration/reports.test.js	Integration	13	13	0	1.5
tests/integration/reservation-entry.test.js	Integration	15	15	0	15.8
tests/integration/reservations.test.js	Integration	15	15	0	10.7
tests/integration/scalability.test.js	Integration	8	8	0	4.1
tests/integration/session-timeout.test.js	Integration	12	12	0	8.0
tests/integration/slot-availability.test.js	Integration	27	27	0	23.9
tests/integration/slots.test.js	Integration	17	17	0	11.4
tests/integration/validation.test.js	Integration	11	11	0	4.9
tests/integration/vehicle-entry.test.js	Integration	15	15	0	1.5
tests/integration/vehicle-exit.test.js	Integration	20	20	0	1.9
tests/integration/vehicles.test.js	Integration	16	16	0	1.4

5. Test Plan cases

Each Jest test carries the Test Plan ID it verifies in its name (for example "UT-08_01: ..."). A case is Pass only when at least one test carries its ID and every such test passed. Results below are computed from this run.

Case ID	Req.	Module	Description	Tests	Result
UT-01_01	FR-01	User Authentication	Valid registration	2	Pass
UT-01_02	FR-01	User Authentication	Valid login	1	Pass
IT-01_03	FR-01	User Authentication	Login with wrong password	1	Pass
IT-01_04	FR-01	User Authentication	Reject duplicate email	1	Pass
IT-01_05	FR-01	User Authentication	Reject invalid email format	2	Pass
ST-01_06	FR-01	User Authentication	Lock account after 5 failed logins	1	Pass
UT-02_01	FR-02	Vehicle Management	Add a new vehicle	2	Pass
UT-02_02	FR-02	Vehicle Management	View registered vehicles	1	Pass
IT-02_03	FR-02	Vehicle Management	Reject duplicate vehicle	1	Pass
IT-02_04	FR-02	Vehicle Management	Reject invalid registration number	2	Pass
IT-02_05	FR-02	Vehicle Management	Update vehicle details	1	Pass
ST-02_06	FR-02	Vehicle Management	Updated vehicle persists after re-login	1	Pass
UT-03_01	FR-03	Parking Availability	Display slot status	1	Pass
UT-03_02	FR-03	Parking Availability	Filter available slots	4	Pass
IT-03_03	FR-03	Parking Availability	Availability after slot is occupied	1	Pass
IT-03_04	FR-03	Parking Availability	Availability after slot is released	1	Pass
IT-03_05	FR-03	Parking Availability	Attendant views availability	3	Pass
ST-03_06	FR-03	Parking Availability	Availability response time	1	Pass
UT-04_01	FR-04	Slot Reservation	Reserve an available slot	2	Pass
UT-04_02	FR-04	Slot Reservation	Reservation requires login	1	Pass

Case ID	Req.	Module	Description	Tests	Result
IT-04_03	FR-04	Slot Reservation	Reject already reserved slot	1	Pass
IT-04_04	FR-04	Slot Reservation	Reject invalid time period	1	Pass
IT-04_05	FR-04	Slot Reservation	Reserved slot not given to another vehicle	1	Pass
ST-04_06	FR-04	Slot Reservation	Reservation shown in occupancy report	1	Pass
UT-05_01	FR-05	Vehicle Entry	Record vehicle entry	1	Pass
UT-05_02	FR-05	Vehicle Entry	Reject unregistered vehicle	1	Pass
IT-05_03	FR-05	Vehicle Entry	Block second active session	1	Pass
IT-05_04	FR-05	Vehicle Entry	Block entry when no slot is free	1	Pass
IT-05_05	FR-05	Vehicle Entry	Entry timestamp is correct	1	Pass
ST-05_06	FR-05	Vehicle Entry	Entry completes within time limit	1	Pass
UT-06_01	FR-06	Slot Assignment	Assign an available slot	1	Pass
UT-06_02	FR-06	Slot Assignment	Assign slot by vehicle type	1	Pass
IT-06_03	FR-06	Slot Assignment	Occupied slot not reassigned	1	Pass
IT-06_04	FR-06	Slot Assignment	Unavailable slot not assigned	1	Pass
IT-06_05	FR-06	Slot Assignment	Attendant overrides slot	0	Fail
ST-06_06	FR-06	Slot Assignment	Slot shows Occupied everywhere	1	Pass
UT-07_01	FR-07	Vehicle Tracking	Find a parked vehicle	1	Pass
UT-07_02	FR-07	Vehicle Tracking	Administrator tracks vehicle	1	Pass
IT-07_03	FR-07	Vehicle Tracking	Vehicle not parked	1	Pass
IT-07_04	FR-07	Vehicle Tracking	Empty vehicle number	3	Pass
IT-07_05	FR-07	Vehicle Tracking	Driver denied vehicle tracking	2	Pass
ST-07_06	FR-07	Vehicle Tracking	Tracking shows latest slot	1	Pass
UT-08_01	FR-08	Vehicle Exit	Record vehicle exit	1	Pass
UT-08_02	FR-08	Vehicle Exit	Slot released after exit	1	Pass
IT-08_03	FR-08	Vehicle Exit	Block exit without active session	1	Pass
IT-08_04	FR-08	Vehicle Exit	Reject exit before entry time	1	Pass
IT-08_05	FR-08	Vehicle Exit	Exit triggers fee calculation	1	Pass
ST-08_06	FR-08	Vehicle Exit	Completed session cannot be edited	1	Pass
UT-09_01	FR-09	Fee Calculation	Fee for whole hours	2	Pass
UT-09_02	FR-09	Fee Calculation	Fee for a partial hour	3	Pass
IT-09_03	FR-09	Fee Calculation	Fee uses configured rate	3	Pass
IT-09_04	FR-09	Fee Calculation	Zero-duration stay	3	Pass
IT-09_05	FR-09	Fee Calculation	Rate change applies to new exits	1	Pass
ST-09_06	FR-09	Fee Calculation	Fee shown without delay	1	Pass
UT-10_01	FR-10	Payment Management	Successful payment	1	Pass
UT-10_02	FR-10	Payment Management	Payment options listed	1	Pass
IT-10_03	FR-10	Payment Management	Failed payment recorded	1	Pass
IT-10_04	FR-10	Payment Management	Reject expired card	1	Pass
IT-10_05	FR-10	Payment Management	Attendant takes payment	1	Pass
ST-10_06	FR-10	Payment Management	Payment linked to session	1	Pass
UT-11_01	FR-11	Parking History	View completed sessions	1	Pass
UT-11_02	FR-11	Parking History	Session record fields	1	Pass
IT-11_03	FR-11	Parking History	Driver cannot edit history	1	Pass
IT-11_04	FR-11	Parking History	Empty history for new user	1	Pass
IT-11_05	FR-11	Parking History	Administrator views a user's history	1	Pass

Case ID	Req.	Module	Description	Tests	Result
ST-11_06	FR-11	Parking History	History with many sessions	1	Pass
UT-12_01	FR-12	Administration and Reports	Generate occupancy report	1	Pass
UT-12_02	FR-12	Administration and Reports	Add or remove a user	3	Pass
IT-12_03	FR-12	Administration and Reports	Only administrators see audit logs	1	Pass
IT-12_04	FR-12	Administration and Reports	Rate change reflects in fee	3	Pass
IT-12_05	FR-12	Administration and Reports	Disable a parking slot	2	Pass
ST-12_06	FR-12	Administration and Reports	Audit log records admin actions	1	Pass

71 Pass, 1 Fail of 72 cases. The one Fail is IT-06_05 (attendant overrides the assigned slot): this was not built, so entry always assigns the first free slot of the right type. It is a documented deviation; an attendant can move a parked vehicle to another slot instead (#102).

6. Tests added during the testing week

- admin-users: an attendant who recorded payments cannot be removed, so payment records are kept (409).
- vehicles.validation: vehicle model over 100 characters and colour over 50 are refused.
- payments.validation: the Luhn check with a doubled digit over 9 (5555 5555 5555 4444).
- reservations.validation: an impossible date with a time zone (2027-13-45T10:00:00Z) is refused.
- lockout: under a minute left reads "1 minute".
- Test Case Document: expected results of IT-03_04, UT-08_01, UT-08_02 and IT-08_05 now say the slot is released and the session completed once the fee is paid (UC-08); ST-07_06 now uses the Move vehicle feature (#102).

Appendix A. Every test and its result

tests/unit/admin.validation.test.js (11/11 passed)

Test	Result	ms
validateUserFilter › empty values mean no filter	Passed	1
validateUserFilter › rejects an unknown role and an over-long search	Passed	0
validateUserFilter › ignores repeated query parameters	Passed	0
validateNewUser › returns clean details with the email normalised (UT-12_02 data)	Passed	1
validateNewUser › requires a known role	Passed	0
validateNewUser › reports registration and role errors together	Passed	0
parseId › reads a positive whole number	Passed	0
parseId › treats "0" as not found with the given message	Passed	4
parseId › treats "1abc" as not found with the given message	Passed	1
parseId › treats "-3" as not found with the given message	Passed	0
parseId › treats "" as not found with the given message	Passed	0

tests/unit/fees.test.js (12/12 passed)

Test	Result	ms
calculateFee (FR-09) › UT-09_01: 10:00 to 12:00 at ₹20/hr costs ₹40	Passed	1
calculateFee (FR-09) › UT-09_02: 10:00 to 10:45 at ₹20/hr bills 45 minutes with a 15-minute step	Passed	0
calculateFee (FR-09) › UT-09_02: the same stay bills a full hour with a 60-minute step	Passed	0
calculateFee (FR-09) › a stay one second into the next step is billed for that step	Passed	0
calculateFee (FR-09) › IT-09_03 data: 1.5 hours at ₹30/hr costs ₹45	Passed	1
calculateFee (FR-09) › IT-09_04 data: a zero-length stay is charged the minimum fee	Passed	0
calculateFee (FR-09) › the minimum fee applies when the time-based fee is lower	Passed	0
calculateFee (FR-09) › a fee that is not whole rupees is rounded up	Passed	0

Test	Result	ms
calculateFee (FR-09) › refuses a duration of -1	Passed	4
calculateFee (FR-09) › refuses a duration of NaN	Passed	0
calculateFee (FR-09) › refuses a duration of Infinity	Passed	1
calculateFee (FR-09) › refuses a duration of undefined	Passed	0

tests/unit/history.validation.test.js (13/13 passed)

Test	Result	ms
validatePageQuery (FR-11) › no paging means the first page of 20	Passed	1
validatePageQuery (FR-11) › reads page and page size	Passed	0
validatePageQuery (FR-11) › rejects page "0"	Passed	0
validatePageQuery (FR-11) › rejects page "-1"	Passed	0
validatePageQuery (FR-11) › rejects page "1.5"	Passed	0
validatePageQuery (FR-11) › rejects page "abc"	Passed	0
validatePageQuery (FR-11) › rejects page "1e3"	Passed	0
validatePageQuery (FR-11) › rejects page " 2"	Passed	0
validatePageQuery (FR-11) › rejects page size "0"	Passed	0
validatePageQuery (FR-11) › rejects page size "101"	Passed	1
validatePageQuery (FR-11) › rejects page size "ten"	Passed	0
validatePageQuery (FR-11) › rejects page size "-5"	Passed	0
validatePageQuery (FR-11) › rejects repeated parameters (arrays)	Passed	0

tests/unit/parking.validation.test.js (12/12 passed)

Test	Result	ms
validateLocateQuery (FR-07) › trims and upper-cases the registration number	Passed	1
validateLocateQuery (FR-07) › removes spaces and hyphens, as vehicles are stored	Passed	0
validateLocateQuery (FR-07) › asks for a registration number when it is undefined	Passed	0
validateLocateQuery (FR-07) › asks for a registration number when it is ""	Passed	0
validateLocateQuery (FR-07) › asks for a registration number when it is " "	Passed	0
validateLocateQuery (FR-07) › asks for a registration number when the query is missing	Passed	0
validateLocateQuery (FR-07) › rejects "KA01<script>"	Passed	0
validateLocateQuery (FR-07) › rejects "KA01_AB"	Passed	1
validateLocateQuery (FR-07) › rejects "KA01;DROP"	Passed	0
validateLocateQuery (FR-07) › rejects "AAAAAAAAAAAAAAAAAAAA"	Passed	0
validateLocateQuery (FR-07) › rejects repeated parameters and other non-text values	Passed	0
validateLocateQuery (FR-07) › the top-level error message is the field message	Passed	0

tests/unit/password.test.js (7/7 passed)

Test	Result	ms
password hashing › stores a bcrypt hash, never the plain password	Passed	55
password hashing › the hash is 60 characters, matching the password_hash CHAR(60) column	Passed	50
password hashing › uses a cost factor of at least 10	Passed	51
password hashing › uses a new random salt every time	Passed	151
password hashing › verifies the right password and rejects a wrong one	Passed	155
password hashing › the stored hash cannot be used as the password	Passed	102
password hashing › rejects any password when there is no stored hash	Passed	51

tests/unit/payments.validation.test.js (24/24 passed)

Test	Result	ms
payment validation (FR-10) › cash needs only the session and method; other details are dropped	Passed	0
payment validation (FR-10) › UPI needs a UPI ID, which is trimmed and lower-cased	Passed	1
payment validation (FR-10) › the UPI ID undefined is rejected	Passed	0

Test	Result	ms
payment validation (FR-10) › the UPI ID "" is rejected	Passed	0
payment validation (FR-10) › the UPI ID "arjun" is rejected	Passed	0
payment validation (FR-10) › the UPI ID "arjun@" is rejected	Passed	0
payment validation (FR-10) › the UPI ID "@okaxis" is rejected	Passed	0
payment validation (FR-10) › the UPI ID "ar jun@okaxis" is rejected	Passed	0
payment validation (FR-10) › the UPI ID "arjun@ok axis" is rejected	Passed	0
payment validation (FR-10) › the UPI ID "arjun@1bank" is rejected	Passed	0
payment validation (FR-10) › the UPI ID 42 is rejected	Passed	0
payment validation (FR-10) › a card number is cleaned of spaces and hyphens	Passed	0
payment validation (FR-10) › the Luhn check catches a mistyped digit	Passed	0
payment validation (FR-10) › the Luhn check subtracts 9 when a doubled digit is over 9	Passed	0
payment validation (FR-10) › a card number that is too short is invalid	Passed	0
payment validation (FR-10) › a card number that is too long is invalid	Passed	0
payment validation (FR-10) › a card number that is letters is invalid	Passed	0
payment validation (FR-10) › a card number that is a number, not text is invalid	Passed	0
payment validation (FR-10) › a card is valid until the end of its expiry month	Passed	1
payment validation (FR-10) › the expiry "13/30" is not MM/YY	Passed	0
payment validation (FR-10) › the expiry "1/30" is not MM/YY	Passed	0
payment validation (FR-10) › the expiry "2030-12" is not MM/YY	Passed	0
payment validation (FR-10) › the expiry "" is not MM/YY	Passed	0
payment validation (FR-10) › an unknown method or a bad session id is rejected	Passed	0

tests/unit/rates.validation.test.js (17/17 passed)

Test	Result	ms
validateRate › accepts the Test Plan rate (IT-12_04 data: ₹25/hr)	Passed	0
validateRate › accepts numbers sent as text, as a form would send them	Passed	1
validateRate › reports every missing field at once	Passed	0
validateRate › rejects an hourly rate of 0	Passed	0
validateRate › rejects an hourly rate of -5	Passed	1
validateRate › rejects an hourly rate of 25.5	Passed	0
validateRate › rejects an hourly rate of "2e1"	Passed	0
validateRate › rejects an hourly rate of "abc"	Passed	0
validateRate › rejects an hourly rate of 10001	Passed	0
validateRate › rejects an hourly rate of true	Passed	1
validateRate › allows a minimum fee of 0 but not a negative one	Passed	0
validateRate › rejects a billing step of 0	Passed	0
validateRate › rejects a billing step of 10	Passed	0
validateRate › rejects a billing step of 45	Passed	0
validateRate › rejects a billing step of "15.0"	Passed	1
parseSlotType › accepts both slot types	Passed	0
parseSlotType › treats anything else as not found	Passed	4

tests/unit/requireRole.test.js (5/5 passed)

Test	Result	ms
requireRole › lets an allowed role through	Passed	1
requireRole › accepts any of several allowed roles	Passed	0
requireRole › refuses a role that is not listed with 403	Passed	0
requireRole › refuses a request with no logged-in user with 401	Passed	1
requireRole › fails fast when given no roles or a misspelt role	Passed	4

tests/unit/reservations.validation.test.js (17/17 passed)

Test	Result	ms
toMysqlDateTime › keeps the local time "2027-01-10T10:00" as "2027-01-10 10:00:00"	Passed	0
toMysqlDateTime › keeps the local time "2027-01-10T10:00:30" as "2027-01-10 10:00:30"	Passed	1
toMysqlDateTime › keeps the local time "2027-01-10 10:00:00" as "2027-01-10 10:00:00"	Passed	0
toMysqlDateTime › converts an ISO time with Z and milliseconds to server local time (the demo bug)	Passed	0
toMysqlDateTime › converts a time with an offset	Passed	0
toMysqlDateTime › refuses "1"	Passed	0
toMysqlDateTime › refuses "March 5"	Passed	0
toMysqlDateTime › refuses "2027-02-30T10:00"	Passed	0
toMysqlDateTime › refuses "2027-13-45T10:00:00Z"	Passed	0
toMysqlDateTime › refuses "2027-01-10T25:00"	Passed	0
toMysqlDateTime › refuses "2027-01-10"	Passed	0
toMysqlDateTime › refuses ""	Passed	0
toMysqlDateTime › refuses undefined	Passed	1
toMysqlDateTime › refuses 1700000000000	Passed	0
validateReservation › returns MySQL-ready times	Passed	0
validateReservation › IT-04_04 data: an end before the start is refused	Passed	0
validateReservation › text that is not a date-time is refused instead of reaching MySQL	Passed	1

tests/unit/slots.filter.validation.test.js (14/14 passed)

Test	Result	ms
validateSlotFilter (FR-03) › no filter at all means no restriction	Passed	1
validateSlotFilter (FR-03) › empty strings mean no restriction	Passed	0
validateSlotFilter (FR-03) › accepts status "available"	Passed	0
validateSlotFilter (FR-03) › accepts status "occupied"	Passed	0
validateSlotFilter (FR-03) › accepts status "reserved"	Passed	1
validateSlotFilter (FR-03) › accepts status "unavailable"	Passed	0
validateSlotFilter (FR-03) › accepts type "two_wheeler"	Passed	0
validateSlotFilter (FR-03) › accepts type "four_wheeler"	Passed	0
validateSlotFilter (FR-03) › rejects status "free"	Passed	1
validateSlotFilter (FR-03) › rejects status "AVAILABLE"	Passed	0
validateSlotFilter (FR-03) › rejects status " available"	Passed	0
validateSlotFilter (FR-03) › rejects status "truck"	Passed	1
validateSlotFilter (FR-03) › rejects an unknown type	Passed	0
validateSlotFilter (FR-03) › rejects repeated parameters (arrays)	Passed	0

tests/unit/slots.validation.test.js (23/23 passed)

Test	Result	ms
validateNewSlot › accepts the Test Plan slot and normalises the code (IT-12_05 data)	Passed	0
validateNewSlot › accepts both slot types	Passed	1
validateNewSlot › reports every missing field at once	Passed	0
validateNewSlot › rejects the code "P 30"	Passed	0
validateNewSlot › rejects the code "P_30"	Passed	0
validateNewSlot › rejects the code "-P30"	Passed	0
validateNewSlot › rejects the code "P30-"	Passed	0
validateNewSlot › rejects the code "P--30"	Passed	0
validateNewSlot › rejects the code "AAAAAAAAAAA"	Passed	0
validateNewSlot › rejects an unknown slot type	Passed	0
validateNewSlot › ignores non-string values	Passed	0
validateSlotUpdate › accepts true and false	Passed	0
validateSlotUpdate › rejects disabled = undefined	Passed	1
validateSlotUpdate › rejects disabled = "true"	Passed	0

Test	Result	ms
validateSlotUpdate › rejects disabled = 1	Passed	0
validateSlotUpdate › rejects disabled = null	Passed	0
parseSlotId › reads a positive whole number	Passed	0
parseSlotId › treats "0" as a missing slot	Passed	4
parseSlotId › treats "-1" as a missing slot	Passed	0
parseSlotId › treats "1abc" as a missing slot	Passed	1
parseSlotId › treats "1.5" as a missing slot	Passed	1
parseSlotId › treats "" as a missing slot	Passed	0
parseSlotId › treats "01" as a missing slot	Passed	0

tests/unit/users.validation.test.js (13/13 passed)

Test	Result	ms
validateRegistration › accepts the Test Plan details and normalises email (UT-01_01 data)	Passed	1
validateRegistration › rejects a malformed email such as abc@com (IT-01_05 data)	Passed	1
validateRegistration › rejects an email longer than 254 characters	Passed	0
validateRegistration › reports every missing field at once	Passed	0
validateRegistration › rejects a name that is too short or too long	Passed	1
validateRegistration › rejects a password shorter than 6 characters	Passed	0
validateRegistration › rejects a password longer than 72 bytes	Passed	0
validateRegistration › counts password length in bytes, not characters	Passed	0
validateRegistration › ignores non-string values	Passed	1
validateLogin › requires email and password	Passed	0
validateLogin › normalises the email	Passed	0
validateLogin › rejects a password longer than 72 bytes	Passed	0
validateLogin › ignores non-string values	Passed	0

tests/unit/vehicles.validation.test.js (21/21 passed)

Test	Result	ms
normaliseRegistration › "ka 01 ab-1234" becomes "KA01AB1234"	Passed	1
normaliseRegistration › " KA-01-AB-1234 " becomes "KA01AB1234"	Passed	0
normaliseRegistration › "KA01AB1234" becomes "KA01AB1234"	Passed	0
normaliseRegistration › non-text values become an empty string	Passed	0
validateVehicle (FR-02) › UT-02_01 data: accepts the Test Plan vehicle and trims the text fields	Passed	0
validateVehicle (FR-02) › accepts both vehicle types	Passed	0
validateVehicle (FR-02) › IT-02_04 data: rejects 1234, which has no letters	Passed	1
validateVehicle (FR-02) › rejects the registration number "KAAB"	Passed	0
validateVehicle (FR-02) › rejects the registration number "KA1"	Passed	0
validateVehicle (FR-02) › rejects the registration number "KA01_AB12"	Passed	0
validateVehicle (FR-02) › rejects the registration number "KA01<b>"	Passed	0
validateVehicle (FR-02) › rejects the registration number "KKKKKKKKKK123456"	Passed	0
validateVehicle (FR-02) › rejects a type that is not a slot type, such as "Car"	Passed	1
validateVehicle (FR-02) › rejects a model over 100 characters and a colour over 50	Passed	0
validateVehicle (FR-02) › reports every missing field at once	Passed	0
parseVehicleId › reads a positive whole number	Passed	0
parseVehicleId › treats "0" as a missing vehicle	Passed	5
parseVehicleId › treats "-1" as a missing vehicle	Passed	0
parseVehicleId › treats "1abc" as a missing vehicle	Passed	1
parseVehicleId › treats "1.5" as a missing vehicle	Passed	0
parseVehicleId › treats "" as a missing vehicle	Passed	0

tests/integration/admin-users.test.js (29/29 passed)

Test	Result	ms
listing users › the admin sees every user with a vehicle count and no password hash	Passed	683
listing users › users can be filtered by role (Bindu's history picker uses role=driver)	Passed	789
listing users › search matches name or email, and % is matched literally	Passed	832
listing users › an unknown role filter is refused with 400	Passed	547
adding and removing users › UT-12_02: the admin adds an attendant, who can log in, and the list updates	Passed	706
adding and removing users › UT-12_02: the admin removes a user, with their vehicles and logins	Passed	802
adding and removing users › an email already in use is refused with 409	Passed	799
adding and removing users › every invalid field is reported at once, including the role	Passed	606
adding and removing users › an admin cannot remove their own account	Passed	588
adding and removing users › a user with parking history is kept, so the history is not lost	Passed	793
adding and removing users › an attendant who recorded payments is kept, so the payment records are not lost	Passed	803
adding and removing users › user id "999999" is not found	Passed	650
adding and removing users › user id "1abc" is not found	Passed	633
managing vehicles › the admin views a user's vehicles	Passed	722
managing vehicles › the admin updates any user's vehicle, with the same rules as the driver	Passed	714
managing vehicles › changing a vehicle to another vehicle's number is refused with 409	Passed	869
managing vehicles › the admin removes a vehicle without history	Passed	802
managing vehicles › a vehicle with parking history is kept	Passed	737
managing vehicles › a driver deleting their own vehicle with history gets 409, not a server error	Passed	734
managing vehicles › vehicle id "999999" is not found	Passed	553
managing vehicles › vehicle id "1abc" is not found	Passed	544
only administrators manage users (NFR-11) › get /api/admin/users without a session gives 401	Passed	544
only administrators manage users (NFR-11) › post /api/admin/users without a session gives 401	Passed	555
only administrators manage users (NFR-11) › delete /api/admin/users/1 without a session gives 401	Passed	548
only administrators manage users (NFR-11) › get /api/admin/users/1/vehicles without a session gives 401	Passed	569
only administrators manage users (NFR-11) › put /api/admin/vehicles/1 without a session gives 401	Passed	529
only administrators manage users (NFR-11) › delete /api/admin/vehicles/1 without a session gives 401	Passed	586
only administrators manage users (NFR-11) › a driver is refused with 403 on every route	Passed	764
only administrators manage users (NFR-11) › a attendant is refused with 403 on every route	Passed	803

tests/integration/audit-log.test.js (10/10 passed)

Test	Result	ms
admin actions are logged (NFR-16) › ST-12_06: a rate update shows the action, the administrator and the time	Passed	805
admin actions are logged (NFR-16) › adding, disabling and enabling a slot are each logged	Passed	559
admin actions are logged (NFR-16) › adding and removing a user, and changing and removing a vehicle, are logged	Passed	794
admin actions are logged (NFR-16) › a refused action is not logged	Passed	570
admin actions are logged (NFR-16) › the entry keeps the admin name after that admin account is removed	Passed	698
reading the log › entries can be filtered by action and paged, newest first	Passed	562
reading the log › an unknown action or bad paging is refused with 400	Passed	469
only administrators read the log (NFR-17) › without a session the request is refused with 401	Passed	524
only administrators read the log (NFR-17) › IT-12_03: an attendant is denied access to the audit log	Passed	659
only administrators read the log (NFR-17) › a driver is denied too	Passed	649

tests/integration/auth.test.js (12/12 passed)

Test	Result	ms
registration › UT-01_01: valid details create an account with a confirmation	Passed	474
registration › the password is stored only as a bcrypt hash	Passed	455
registration › IT-01_04: a duplicate email is rejected	Passed	578
registration › IT-01_05: a malformed email (abc@com) is rejected	Passed	446
registration › a new account is always a driver, even if another role is sent	Passed	505
login › UT-01_02: correct credentials log in and open a session	Passed	583

Test	Result	ms
login › IT-01_03: a wrong password shows an error and denies access	Passed	558
login › an unknown email gets the same error as a wrong password	Passed	524
login › only a hash of the session token is stored	Passed	569
session › protected requests without a session are refused	Passed	442
session › a made-up session token is refused	Passed	398
session › logout ends the session	Passed	603

tests/integration/authorization.test.js (15/15 passed)

Test	Result	ms
NFR-10: protected functions need a login › /sample/driver without a session is refused with 401	Passed	3
NFR-10: protected functions need a login › /sample/staff without a session is refused with 401	Passed	2
NFR-10: protected functions need a login › /sample/admin without a session is refused with 401	Passed	1
NFR-10: protected functions need a login › /api/auth/me without a session is refused with 401	Passed	2
NFR-11: each role has distinct access › driver → /sample/driver gives 200	Passed	4
NFR-11: each role has distinct access › driver → /sample/staff gives 403	Passed	3
NFR-11: each role has distinct access › driver → /sample/admin gives 403	Passed	3
NFR-11: each role has distinct access › attendant → /sample/driver gives 403	Passed	3
NFR-11: each role has distinct access › attendant → /sample/staff gives 200	Passed	3
NFR-11: each role has distinct access › attendant → /sample/admin gives 403	Passed	3
NFR-11: each role has distinct access › admin → /sample/driver gives 403	Passed	2
NFR-11: each role has distinct access › admin → /sample/staff gives 200	Passed	3
NFR-11: each role has distinct access › admin → /sample/admin gives 200	Passed	3
NFR-11: each role has distinct access › IT-07_05: a driver trying a staff-only action (track vehicle) is denied	Passed	3
NFR-11: each role has distinct access › every role can still see its own profile	Passed	8

tests/integration/data-protection.test.js (157/157 passed)

Test	Result	ms
every route says who may call it › the list above matches the routes in the server code	Passed	7
every route says who may call it › GET /api/auth/me without a login gives 401	Passed	2
every route says who may call it › GET /api/vehicles without a login gives 401	Passed	3
every route says who may call it › GET /api/vehicles/:id without a login gives 401	Passed	2
every route says who may call it › POST /api/vehicles without a login gives 401	Passed	2
every route says who may call it › PUT /api/vehicles/:id without a login gives 401	Passed	2
every route says who may call it › DELETE /api/vehicles/:id without a login gives 401	Passed	2
every route says who may call it › GET /api/slots without a login gives 401	Passed	1
every route says who may call it › POST /api/slots without a login gives 401	Passed	2
every route says who may call it › PATCH /api/slots/:id without a login gives 401	Passed	2
every route says who may call it › GET /api/reservations without a login gives 401	Passed	2
every route says who may call it › POST /api/reservations without a login gives 401	Passed	3
every route says who may call it › POST /api/parking/assign without a login gives 401	Passed	1
every route says who may call it › POST /api/parking/entry without a login gives 401	Passed	2
every route says who may call it › POST /api/parking/exit without a login gives 401	Passed	2
every route says who may call it › POST /api/parking/move without a login gives 401	Passed	2
every route says who may call it › GET /api/parking/locate without a login gives 401	Passed	2
every route says who may call it › GET /api/rates without a login gives 401	Passed	2
every route says who may call it › PUT /api/rates/:type without a login gives 401	Passed	1
every route says who may call it › GET /api/admin/users without a login gives 401	Passed	2
every route says who may call it › POST /api/admin/users without a login gives 401	Passed	2
every route says who may call it › DELETE /api/admin/users/:id without a login gives 401	Passed	2
every route says who may call it › GET /api/admin/users/:id/vehicles without a login gives 401	Passed	1
every route says who may call it › PUT /api/admin/vehicles/:id without a login gives 401	Passed	2
every route says who may call it › DELETE /api/admin/vehicles/:id without a login gives 401	Passed	1

Test	Result	ms
every route says who may call it › GET /api/history without a login gives 401	Passed	1
every route says who may call it › GET /api/history/users/:id without a login gives 401	Passed	2
every route says who may call it › GET /api/payments/methods without a login gives 401	Passed	1
every route says who may call it › POST /api/payments without a login gives 401	Passed	2
every route says who may call it › GET /api/payments/unpaid without a login gives 401	Passed	2
every route says who may call it › GET /api/payments/awaiting without a login gives 401	Passed	2
every route says who may call it › GET /api/audit without a login gives 401	Passed	2
every route says who may call it › GET /api/reports/occupancy without a login gives 401	Passed	1
every route says who may call it › GET /api/vehicles is refused for a attendant with 403	Passed	4
every route says who may call it › GET /api/vehicles is refused for a admin with 403	Passed	3
every route says who may call it › GET /api/vehicles/:id is refused for a attendant with 403	Passed	3
every route says who may call it › GET /api/vehicles/:id is refused for a admin with 403	Passed	3
every route says who may call it › POST /api/vehicles is refused for a attendant with 403	Passed	3
every route says who may call it › POST /api/vehicles is refused for a admin with 403	Passed	3
every route says who may call it › PUT /api/vehicles/:id is refused for a attendant with 403	Passed	2
every route says who may call it › PUT /api/vehicles/:id is refused for a admin with 403	Passed	3
every route says who may call it › DELETE /api/vehicles/:id is refused for a attendant with 403	Passed	3
every route says who may call it › DELETE /api/vehicles/:id is refused for a admin with 403	Passed	2
every route says who may call it › POST /api/slots is refused for a driver with 403	Passed	3
every route says who may call it › POST /api/slots is refused for a attendant with 403	Passed	3
every route says who may call it › PATCH /api/slots/:id is refused for a driver with 403	Passed	3
every route says who may call it › PATCH /api/slots/:id is refused for a attendant with 403	Passed	3
every route says who may call it › POST /api/parking/assign is refused for a admin with 403	Passed	4
every route says who may call it › POST /api/parking/entry is refused for a driver with 403	Passed	3
every route says who may call it › POST /api/parking/entry is refused for a admin with 403	Passed	3
every route says who may call it › POST /api/parking/exit is refused for a driver with 403	Passed	4
every route says who may call it › POST /api/parking/move is refused for a driver with 403	Passed	3
every route says who may call it › POST /api/parking/move is refused for a admin with 403	Passed	3
every route says who may call it › GET /api/parking/locate is refused for a driver with 403	Passed	3
every route says who may call it › GET /api/rates is refused for a driver with 403	Passed	3
every route says who may call it › GET /api/rates is refused for a attendant with 403	Passed	3
every route says who may call it › PUT /api/rates/:type is refused for a driver with 403	Passed	3
every route says who may call it › PUT /api/rates/:type is refused for a attendant with 403	Passed	2
every route says who may call it › GET /api/admin/users is refused for a driver with 403	Passed	2
every route says who may call it › GET /api/admin/users is refused for a attendant with 403	Passed	3
every route says who may call it › POST /api/admin/users is refused for a driver with 403	Passed	2
every route says who may call it › POST /api/admin/users is refused for a attendant with 403	Passed	2
every route says who may call it › DELETE /api/admin/users/:id is refused for a driver with 403	Passed	3
every route says who may call it › DELETE /api/admin/users/:id is refused for a attendant with 403	Passed	3
every route says who may call it › GET /api/admin/users/:id/vehicles is refused for a driver with 403	Passed	3
every route says who may call it › GET /api/admin/users/:id/vehicles is refused for a attendant with 403	Passed	2
every route says who may call it › PUT /api/admin/vehicles/:id is refused for a driver with 403	Passed	3
every route says who may call it › PUT /api/admin/vehicles/:id is refused for a attendant with 403	Passed	2
every route says who may call it › DELETE /api/admin/vehicles/:id is refused for a driver with 403	Passed	3
every route says who may call it › DELETE /api/admin/vehicles/:id is refused for a attendant with 403	Passed	2
every route says who may call it › GET /api/history is refused for a attendant with 403	Passed	2
every route says who may call it › GET /api/history is refused for a admin with 403	Passed	3
every route says who may call it › GET /api/history/users/:id is refused for a driver with 403	Passed	3
every route says who may call it › GET /api/history/users/:id is refused for a attendant with 403	Passed	3
every route says who may call it › GET /api/payments/unpaid is refused for a attendant with 403	Passed	4
every route says who may call it › GET /api/payments/unpaid is refused for a admin with 403	Passed	3
every route says who may call it › GET /api/payments/awaiting is refused for a driver with 403	Passed	3
every route says who may call it › GET /api/audit is refused for a driver with 403	Passed	2
every route says who may call it › GET /api/audit is refused for a attendant with 403	Passed	3

Test	Result	ms
every route says who may call it › GET /api/reports/occupancy is refused for a driver with 403	Passed	4
every route says who may call it › GET /api/reports/occupancy is refused for a attendant with 403	Passed	3
every route says who may call it › GET /api/auth/me is open to a driver (passes the access check)	Passed	2
every route says who may call it › GET /api/auth/me is open to a attendant (passes the access check)	Passed	3
every route says who may call it › GET /api/auth/me is open to a admin (passes the access check)	Passed	2
every route says who may call it › GET /api/vehicles is open to a driver (passes the access check)	Passed	4
every route says who may call it › GET /api/vehicles/:id is open to a driver (passes the access check)	Passed	3
every route says who may call it › POST /api/vehicles is open to a driver (passes the access check)	Passed	3
every route says who may call it › PUT /api/vehicles/:id is open to a driver (passes the access check)	Passed	2
every route says who may call it › DELETE /api/vehicles/:id is open to a driver (passes the access check)	Passed	3
every route says who may call it › GET /api/slots is open to a driver (passes the access check)	Passed	4
every route says who may call it › GET /api/slots is open to a attendant (passes the access check)	Passed	3
every route says who may call it › GET /api/slots is open to a admin (passes the access check)	Passed	2
every route says who may call it › POST /api/slots is open to a admin (passes the access check)	Passed	3
every route says who may call it › PATCH /api/slots/:id is open to a admin (passes the access check)	Passed	2
every route says who may call it › GET /api/reservations is open to a driver (passes the access check)	Passed	3
every route says who may call it › GET /api/reservations is open to a attendant (passes the access check)	Passed	3
every route says who may call it › GET /api/reservations is open to a admin (passes the access check)	Passed	3
every route says who may call it › POST /api/reservations is open to a driver (passes the access check)	Passed	3
every route says who may call it › POST /api/reservations is open to a attendant (passes the access check)	Passed	2
every route says who may call it › POST /api/reservations is open to a admin (passes the access check)	Passed	3
every route says who may call it › POST /api/parking/assign is open to a driver (passes the access check)	Passed	2
every route says who may call it › POST /api/parking/assign is open to a attendant (passes the access check)	Passed	3
every route says who may call it › POST /api/parking/entry is open to a attendant (passes the access check)	Passed	4
every route says who may call it › POST /api/parking/exit is open to a attendant (passes the access check)	Passed	3
every route says who may call it › POST /api/parking/exit is open to a admin (passes the access check)	Passed	3
every route says who may call it › POST /api/parking/move is open to a attendant (passes the access check)	Passed	4
every route says who may call it › GET /api/parking/locate is open to a attendant (passes the access check)	Passed	2
every route says who may call it › GET /api/parking/locate is open to a admin (passes the access check)	Passed	10
every route says who may call it › GET /api/rates is open to a admin (passes the access check)	Passed	18
every route says who may call it › PUT /api/rates/:type is open to a admin (passes the access check)	Passed	4
every route says who may call it › GET /api/admin/users is open to a admin (passes the access check)	Passed	3
every route says who may call it › POST /api/admin/users is open to a admin (passes the access check)	Passed	4
every route says who may call it › DELETE /api/admin/users/:id is open to a admin (passes the access check)	Passed	2
every route says who may call it › GET /api/admin/users/:id/vehicles is open to a admin (passes the access check)	Passed	2
every route says who may call it › PUT /api/admin/vehicles/:id is open to a admin (passes the access check)	Passed	3
every route says who may call it › DELETE /api/admin/vehicles/:id is open to a admin (passes the access check)	Passed	3
every route says who may call it › GET /api/history is open to a driver (passes the access check)	Passed	4
every route says who may call it › GET /api/history/users/:id is open to a admin (passes the access check)	Passed	4
every route says who may call it › GET /api/payments/methods is open to a driver (passes the access check)	Passed	3
every route says who may call it › GET /api/payments/methods is open to a attendant (passes the access check)	Passed	3
every route says who may call it › GET /api/payments/methods is open to a admin (passes the access check)	Passed	3
every route says who may call it › POST /api/payments is open to a driver (passes the access check)	Passed	3
every route says who may call it › POST /api/payments is open to a attendant (passes the access check)	Passed	3
every route says who may call it › POST /api/payments is open to a admin (passes the access check)	Passed	2
every route says who may call it › GET /api/payments/unpaid is open to a driver (passes the access check)	Passed	4
every route says who may call it › GET /api/payments/awaiting is open to a attendant (passes the access check)	Passed	3
every route says who may call it › GET /api/payments/awaiting is open to a admin (passes the access check)	Passed	3
every route says who may call it › GET /api/audit is open to a admin (passes the access check)	Passed	4
every route says who may call it › GET /api/reports/occupancy is open to a admin (passes the access check)	Passed	2
every route says who may call it › registration, login and logout need no login	Passed	5
one user cannot reach another user's data › another driver cannot read, change or delete a driver's vehicle	Passed	12
one user cannot reach another user's data › a driver's vehicle list holds only their own vehicles	Passed	3
one user cannot reach another user's data › another driver's vehicle cannot be parked or reserved by someone else	Passed	7

Test	Result	ms
one user cannot reach another user's data › another driver cannot pay for a driver's session, and gets the same answer as for a missing one	Passed	8
one user cannot reach another user's data › history, unpaid sessions and reservations show only the logged-in driver's own	Passed	9
one user cannot reach another user's data › a driver cannot open another user's history, and an attendant cannot open any history or a driver's unpaid list	Passed	10
one user cannot reach another user's data › only an administrator can list users, read a user's vehicles or read a user's history	Passed	19
one user cannot reach another user's data › the audit log and the reports are for administrators only	Passed	18
confidential values never come back, and are never stored or logged › no response holds a password hash, a lockout field, a session token or card details	Passed	0
confidential values never come back, and are never stored or logged › the payment response shows only the last 4 card digits	Passed	9
confidential values never come back, and are never stored or logged › the database keeps only the last 4 card digits and never the expiry	Passed	5
confidential values never come back, and are never stored or logged › the audit log records who did what, but never a password	Passed	4
confidential values never come back, and are never stored or logged › the password, session token and card details are not in the server logs	Passed	1
confidential values never come back, and are never stored or logged › a database error does not write the SQL, with its values, to the log	Passed	59
private responses are marked so browsers and proxies do not keep them › a data response is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › the profile is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a list of users is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › the audit log is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a report is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a reservations list is not cacheable and carries the security headers	Passed	3
private responses are marked so browsers and proxies do not keep them › a refusal (401) is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a refusal (403) is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a missing route (404) is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a validation error (400) is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › a request with broken JSON (400) is not cacheable and carries the security headers	Passed	4
private responses are marked so browsers and proxies do not keep them › the health check is not cacheable and carries the security headers	Passed	2
private responses are marked so browsers and proxies do not keep them › the session cookie cannot be read by page scripts or sent from other sites	Passed	59

tests/integration/exit-awaiting-payment.test.js (5/5 passed)

Test	Result	ms
UPI fails at the exit, the driver pays cash later › the whole visit: exit, failed UPI, wait, cash, slot released, vehicle can come back	Passed	72
UPI fails at the exit, the driver pays cash later › a vehicle that has not paid cannot enter again	Passed	33
UPI fails at the exit, the driver pays cash later › a slot held by a vehicle waiting to pay is not given to another vehicle	Passed	36
UPI fails at the exit, the driver pays cash later › the driver can also pay from their own Payments page while waiting	Passed	33
UPI fails at the exit, the driver pays cash later › visitors cannot see who is waiting to pay	Passed	8

tests/integration/locate-vehicle.test.js (18/18 passed)

Test	Result	ms
searching for a parked vehicle › UT-07_01: an attendant gets the vehicle’s slot and status	Passed	967
searching for a parked vehicle › UT-07_02: an administrator can search too	Passed	957
searching for a parked vehicle › the search ignores case and surrounding spaces	Passed	858
searching for a parked vehicle › the answer is never cached	Passed	876
searching for a parked vehicle › the result shows the slot’s current status, e.g. Unavailable if an admin disabled it	Passed	959

Test	Result	ms
a vehicle that is not parked › IT-07_03: a registered vehicle with no session returns "no active session found"	Passed	869
a vehicle that is not parked › a vehicle whose session is completed returns "no active session found"	Passed	952
a vehicle that is not parked › an unknown registration number gets the same answer	Passed	856
invalid input › IT-07_04: an empty search (undefined) shows a validation message	Passed	893
invalid input › IT-07_04: an empty search ("") shows a validation message	Passed	821
invalid input › IT-07_04: an empty search (" ") shows a validation message	Passed	848
invalid input › an invalid search ("KA01<script>") shows a validation message	Passed	883
invalid input › an invalid search ("KA01_AB") shows a validation message	Passed	892
invalid input › an invalid search ("AAAAAAAAAAAAAAAAAAAA") shows a validation message	Passed	883
invalid input › a repeated parameter is refused with 400	Passed	948
access (NFR-11) › IT-07_05: a driver trying to locate a vehicle is denied	Passed	869
access (NFR-11) › without a session the search is refused with 401	Passed	781
latest slot after reassignment › ST-07_06: the result shows the new slot after the session is moved	Passed	915

tests/integration/lockout.test.js (11/11 passed)

Test	Result	ms
locking › 4 failed logins do not lock the account	Passed	897
locking › ST-01_06: the 5th consecutive failed login locks the account with a clear message	Passed	842
locking › a locked account refuses even the correct password and opens no session	Passed	817
locking › the message shows the remaining time	Passed	825
locking › under a minute left reads "1 minute", not "1 minutes" or "0 minutes"	Passed	840
locking › locking one account does not affect another	Passed	1001
locking › an unknown email is never locked and always gets the usual error	Passed	815
recovery › a successful login resets the failed-attempt counter	Passed	1354
recovery › the lock ends when the lock period is over	Passed	951
recovery › after a lock ends the user gets a fresh set of 5 attempts	Passed	1195
response › lockout fields are not returned to the client	Passed	665

tests/integration/move-vehicle.test.js (30/30 passed)

Test	Result	ms
moving a parked vehicle › the vehicle is in the new slot, and both slots show it straight away	Passed	968
moving a parked vehicle › the registration number is normalised like everywhere else	Passed	861
moving a parked vehicle › the same session continues, with the entry time unchanged	Passed	903
moving a parked vehicle › the fee is the same as if the vehicle had not moved, and paying frees the new slot	Passed	926
moving a parked vehicle › the old slot can be given to the next vehicle	Passed	894
only a free slot of the vehicle's type can be chosen › an occupied slot is refused (DR-03)	Passed	846
only a free slot of the vehicle's type can be chosen › a slot waiting for payment is still occupied (UC-08)	Passed	924
only a free slot of the vehicle's type can be chosen › a disabled slot is refused	Passed	886
only a free slot of the vehicle's type can be chosen › a slot reserved for the current time is refused (DR-08)	Passed	897
only a free slot of the vehicle's type can be chosen › a slot of another type is refused (DR-07)	Passed	925
only a free slot of the vehicle's type can be chosen › the slot the vehicle is already in is refused	Passed	964
only a free slot of the vehicle's type can be chosen › a slot that does not exist is refused	Passed	947
a vehicle that cannot be moved › one that is not parked is refused	Passed	911
a vehicle that cannot be moved › an unknown registration number gets the same answer	Passed	876
a vehicle that cannot be moved › one waiting to pay is refused and stays in its slot	Passed	850
a vehicle that cannot be moved › one whose stay is over and paid is refused, so a past record cannot change (DR-10)	Passed	883
invalid input › a slot of undefined is refused with a field error	Passed	924
invalid input › a slot of null is refused with a field error	Passed	879
invalid input › a slot of "" is refused with a field error	Passed	927
invalid input › a slot of "abc" is refused with a field error	Passed	888
invalid input › a slot of 0 is refused with a field error	Passed	847
invalid input › a slot of -1 is refused with a field error	Passed	829

Test	Result	ms
invalid input › a slot of 1.5 is refused with a field error	Passed	799
invalid input › a slot of "2" is refused with a field error	Passed	862
invalid input › a missing registration number is refused with a field error	Passed	841
two requests at the same time (NFR-04) › two vehicles moving to the same slot cannot both get it	Passed	876
two requests at the same time (NFR-04) › a move and an entry cannot both get the same slot	Passed	872
access › a driver is refused with 403	Passed	829
access › an administrator is refused with 403	Passed	829
access › without a session the move is refused with 401	Passed	808

tests/integration/parking-history.test.js (29/29 passed)

Test	Result	ms
a driver views their own history › UT-11_01: past completed sessions are listed with their fee and status, newest first	Passed	1107
a driver views their own history › UT-11_02: each record shows vehicle, slot, entry, exit, duration, fee and status	Passed	931
a driver views their own history › a stay recorded through entry, exit and payment appears in history with the fee shown on exit	Passed	939
a driver views their own history › a stay that has exited but is still waiting for payment is not history yet	Passed	914
a driver views their own history › a stay that has not ended yet is not history	Passed	937
a driver views their own history › another driver's sessions are never shown	Passed	915
a driver views their own history › history keeps the vehicle after it is no longer parked, with no password or other user data	Passed	964
past sessions cannot be edited (DR-10) › IT-11_03: no route changes a session, and the record stays as it was	Passed	955
past sessions cannot be edited (DR-10) › a completed session cannot be exited again to change its fee	Passed	889
a new user › IT-11_04: a user with no sessions gets an empty history, not an error	Passed	943
an administrator views any user’s history › IT-11_05: the admin opens a driver’s history	Passed	881
an administrator views any user’s history › a user with no sessions shows an empty history to the admin	Passed	1027
an administrator views any user’s history › user id "999999" is not found	Passed	945
an administrator views any user’s history › user id "1abc" is not found	Passed	963
an administrator views any user’s history › user id "0" is not found	Passed	1022
a user with 50+ sessions › ST-11_06: paging through reaches every session once, without errors	Passed	914
a user with 50+ sessions › pages hold 20, 20 and 15 sessions, newest first	Passed	987
a user with 50+ sessions › all sessions can be loaded in one request with a larger page size	Passed	947
a user with 50+ sessions › the total fee covers every session, not just the current page	Passed	955
a user with 50+ sessions › a page past the end is empty, not an error	Passed	959
a user with 50+ sessions › the admin can page through the same history	Passed	933
input and access (NFR-11, NFR-15) › paging {"page": "0"} is refused with 400	Passed	939
input and access (NFR-11, NFR-15) › paging {"page": "abc"} is refused with 400	Passed	940
input and access (NFR-11, NFR-15) › paging {"pageSize": "101"} is refused with 400	Passed	958
input and access (NFR-11, NFR-15) › paging {"pageSize": "0"} is refused with 400	Passed	950
input and access (NFR-11, NFR-15) › without a session history is refused with 401	Passed	985
input and access (NFR-11, NFR-15) › an attendant is refused with 403 on both routes	Passed	956
input and access (NFR-11, NFR-15) › a driver cannot open another user’s history	Passed	916
input and access (NFR-11, NFR-15) › the admin uses the per-user route, so /api/history itself is driver-only	Passed	959

tests/integration/parking.test.js (13/13 passed)

Test	Result	ms
automatic parking slot assignment (Issue #62) › UT-06_01: assigns an available slot matching the vehicle type	Passed	932
automatic parking slot assignment (Issue #62) › returns 409 when no suitable slot is available	Passed	881
automatic parking slot assignment (Issue #62) › IT-06_04: does not assign a disabled slot	Passed	888
automatic parking slot assignment (Issue #62) › IT-06_03: does not assign a slot with an active parking session	Passed	851
automatic parking slot assignment (Issue #62) › does not allow the same vehicle to have two active sessions	Passed	933
automatic parking slot assignment (Issue #62) › returns 404 for a vehicle that does not belong to the driver	Passed	1054
automatic parking slot assignment (Issue #62) › requires authentication	Passed	896
automatic parking slot assignment (Issue #62) › rejects an invalid vehicle ID	Passed	868
record vehicle entry (Issue #60) › allows an attendant to record entry for another driver vehicle	Passed	995

Test	Result	ms
record vehicle entry (Issue #60) › allows an attendant to record entry for a registered vehicle	Passed	805
record vehicle entry (Issue #60) › records an entry timestamp for the parking session	Passed	771
record vehicle entry (Issue #60) › rejects an unregistered vehicle	Passed	797
record vehicle entry (Issue #60) › records vehicle entry within three seconds	Passed	815

tests/integration/password-security.test.js (7/7 passed)

Test	Result	ms
storage › registration stores only a 60-character bcrypt hash	Passed	474
storage › the plaintext password appears in no column of the users table	Passed	450
storage › two accounts with the same password get different hashes	Passed	566
verification at login › the correct password is accepted and a wrong one refused	Passed	604
verification at login › the stored hash cannot be used to log in	Passed	544
no leaks › no response contains the password or its hash	Passed	483
no leaks › the password is never written to the logs	Passed	637

tests/integration/payments.test.js (21/21 passed)

Test	Result	ms
making a payment (FR-10) › UT-10_01: a successful UPI payment of ₹40 is recorded and the session is marked Paid	Passed	27
making a payment (FR-10) › UT-10_02: the available payment methods are listed	Passed	16
making a payment (FR-10) › IT-10_03: a declined card is recorded as Failed, the user is told, and the session stays unpaid	Passed	25
making a payment (FR-10) › IT-10_04: an expired or invalid card is rejected and nothing is recorded	Passed	32
making a payment (FR-10) › IT-10_05: the attendant takes ₹40 in cash on the driver's behalf	Passed	28
making a payment (FR-10) › ST-10_06: the payment is linked to the correct session, for the session's own fee (SYS-13)	Passed	44
payment rules › a driver cannot pay for another driver's session	Passed	24
payment rules › a paid session cannot be paid again	Passed	27
payment rules › two payments at the same moment record only one	Passed	27
payment rules › a vehicle that is still parked cannot be paid for yet	Passed	21
payment rules › a UPI payment needs a valid UPI ID, and nothing is recorded without one	Passed	37
payment rules › the UPI ID is never stored (NFR-18)	Passed	35
payment rules › a driver cannot choose cash	Passed	21
payment rules › a missing session or method is rejected	Passed	11
payment rules › the stored payment never contains the full card number or the expiry (NFR-18)	Passed	18
unpaid sessions › a driver sees only their own sessions that are waiting for payment	Passed	59
unpaid sessions › only drivers have an unpaid list	Passed	14
a failed UPI payment, then cash later (UC-08) › a declined UPI payment is recorded as failed and the session keeps waiting	Passed	239
a failed UPI payment, then cash later (UC-08) › the attendant finds the waiting vehicle and takes cash when the driver comes back	Passed	53
a failed UPI payment, then cash later (UC-08) › the waiting list shows the longest wait first and leaves out parked and paid sessions	Passed	66
a failed UPI payment, then cash later (UC-08) › only staff see the waiting list	Passed	12

tests/integration/rates.test.js (15/15 passed)

Test	Result	ms
viewing rates › the admin sees the rate for each vehicle type	Passed	66
changing a rate › the new rate is saved and returned	Passed	71
changing a rate › changing one vehicle type leaves the other alone	Passed	75
changing a rate › a rate that is missing is created	Passed	55
changing a rate › invalid values are refused with 400 and the rate is unchanged	Passed	52
changing a rate › an unknown vehicle type is not found	Passed	54
fees use the configured rate (FR-09) › IT-12_04: after a rate change, the next fee uses the new rate	Passed	60
fees use the configured rate (FR-09) › IT-09_03 data: a 1.5-hour stay at ₹30/hr costs ₹45	Passed	61
fees use the configured rate (FR-09) › IT-09_04 data: a zero-length stay is charged the minimum fee	Passed	44

Test	Result	ms
fees use the configured rate (FR-09) › each vehicle type is charged its own rate	Passed	40
fees use the configured rate (FR-09) › a fee cannot be worked out when no rate is set	Passed	100
only administrators set rates (NFR-11) › get /api/rates without a session gives 401	Passed	50
only administrators set rates (NFR-11) › put /api/rates/four_wheeler without a session gives 401	Passed	48
only administrators set rates (NFR-11) › a driver is refused with 403 and the rate is unchanged	Passed	160
only administrators set rates (NFR-11) › a attendant is refused with 403 and the rate is unchanged	Passed	172

tests/integration/reliability.test.js (18/18 passed)

Test	Result	ms
availability (NFR-03) › the health check reports the application and the database as up	Passed	2
availability (NFR-03) › the health check says down (503) when the database does not answer	Passed	1
availability (NFR-03) › a database error gives a clean 500 and the next request works again	Passed	12
availability (NFR-03) › 99% of 600 mixed requests at 20 at a time are answered without a server error	Passed	970
simultaneous operations keep slot and session data consistent (NFR-06) › the same vehicle entering 10 times at once gets exactly one session	Passed	29
simultaneous operations keep slot and session data consistent (NFR-06) › 30 bikes entering at once for 10 slots: exactly 10 get a slot and no slot is given twice	Passed	91
simultaneous operations keep slot and session data consistent (NFR-06) › the same vehicle exiting 10 times at once records one exit, with one fee	Passed	34
simultaneous operations keep slot and session data consistent (NFR-06) › exit finishes even when every other database connection is busy (no pool starvation)	Passed	24
simultaneous operations keep slot and session data consistent (NFR-06) › 3 rounds of bikes entering while others exit never double-book a slot or return a server error	Passed	361
simultaneous operations keep slot and session data consistent (NFR-06) › cars and bikes entering together each get a slot of their own type	Passed	193
simultaneous operations keep slot and session data consistent (NFR-06) › the slot list agrees with the sessions: occupied slots = active sessions	Passed	79
relationships stay accurate (NFR-09) › a stay leaves a complete record: vehicle, slot, times, fee, then paid and completed	Passed	30
relationships stay accurate (NFR-09) › the checker finds a slot with two vehicles in it when one is planted	Passed	19
relationships stay accurate (NFR-09) › the checker finds a wrong fee when one is planted	Passed	25
relationships stay accurate (NFR-09) › a session cannot point to a vehicle or slot that does not exist	Passed	7
relationships stay accurate (NFR-09) › a vehicle or slot with parking history cannot be deleted	Passed	20
relationships stay accurate (NFR-09) › removing a user without history removes their vehicles and logins and leaves no orphans	Passed	19
relationships stay accurate (NFR-09) › removing a user with parking history is refused, so the history stays linked	Passed	18

tests/integration/reports.test.js (13/13 passed)

Test	Result	ms
current occupancy › UT-12_01: a report for today shows the occupancy right now	Passed	23
current occupancy › ST-04_06: a reservation covering now counts in the occupancy	Passed	17
the chosen dates › totals cover the range, with the end date included in full	Passed	57
the chosen dates › every day in the range has a row, including empty days	Passed	62
the chosen dates › a range with no activity reports zeros and no average stay	Passed	58
checking the dates › from="" to="2026-10-12" is refused	Passed	20
checking the dates › from="2026-02-30" to="2026-03-01" is refused	Passed	6
checking the dates › from="2026-10-12" to="2026-10-10" is refused	Passed	6
checking the dates › from="2025-01-01" to="2026-10-10" is refused	Passed	5
checking the dates › from="10/10/2026" to="tomorrow" is refused	Passed	5
only administrators see reports (NFR-11) › without a session the request is refused with 401	Passed	4
only administrators see reports (NFR-11) › a driver is refused with 403	Passed	117
only administrators see reports (NFR-11) › a attendant is refused with 403	Passed	122

tests/integration/reservation-entry.test.js (15/15 passed)

Test	Result	ms
the vehicle with the reservation arrives › it enters when its reserved slot is the only one left	Passed	1031

Test	Result	ms
the vehicle with the reservation arrives › it goes to the slot it reserved, not the first free one	Passed	1027
the vehicle with the reservation arrives › a reservation made through the API works the same way	Passed	1017
the vehicle with the reservation arrives › the reservation is marked completed	Passed	968
the vehicle with the reservation arrives › once the stay is paid the slot is Available, even if the reserved time has not ended	Passed	1061
the vehicle with the reservation arrives › the attendant entry and the driver entry behave the same	Passed	1029
the reserved slot cannot take the vehicle › still occupied: it gets another free slot and the reservation is used	Passed	1048
the reserved slot cannot take the vehicle › disabled by an admin: it gets another free slot	Passed	1045
the reserved slot cannot take the vehicle › occupied and no other slot free: refused, and the reservation is kept	Passed	1016
the reserved slot cannot take the vehicle › wrong vehicle type: it is not parked in the reserved slot (DR-07)	Passed	1006
the reservation still holds the slot against everything else › another vehicle is refused the reserved slot (DR-08)	Passed	1016
the reservation still holds the slot against everything else › another vehicle is sent to a different slot, and the holder still gets theirs	Passed	1087
the reservation still holds the slot against everything else › a reservation for later is neither used nor completed by entering now	Passed	1126
the reservation still holds the slot against everything else › a reservation that has ended is ignored	Passed	1081
the reservation still holds the slot against everything else › a cancelled reservation is ignored	Passed	963

tests/integration/reservations.test.js (15/15 passed)

Test	Result	ms
POST /api/reservations › an ISO time with Z and milliseconds is accepted, not a server error (Sprint 1 demo bug)	Passed	735
POST /api/reservations › text that is not a date-time is refused with 400	Passed	673
POST /api/reservations › UT-04_01: creates a reservation for the logged-in driver	Passed	689
POST /api/reservations › UT-04_02: requires login	Passed	679
POST /api/reservations › rejects another driver's vehicle	Passed	684
POST /api/reservations › rejects a disabled slot	Passed	625
POST /api/reservations › rejects a missing slot	Passed	632
POST /api/reservations › IT-04_03: rejects overlapping reservations for the same slot	Passed	704
POST /api/reservations › allows non-overlapping reservations for the same slot	Passed	643
POST /api/reservations › rejects an end time before the start time	Passed	674
POST /api/reservations › rejects an end time equal to the start time	Passed	702
a reservation holds its slot › UT-04_01: the slot shows Reserved during the reservation	Passed	664
a reservation holds its slot › IT-04_05: a reserved slot is not given to another vehicle (DR-08)	Passed	879
GET /api/reservations › requires login	Passed	678
GET /api/reservations › returns only the logged-in driver's reservations	Passed	780

tests/integration/scalability.test.js (8/8 passed)

Test	Result	ms
test data volume › the database holds 500 slots and 5,000 vehicles	Passed	2
100 concurrent users › all 100 users log in at the same time and get a session	Passed	1
100 concurrent users › all 100 users view the 500-slot list at once, 3 rounds, p95 under 2 seconds	Passed	594
100 concurrent users › all 100 users load their own 50 vehicles at once, p95 under 2 seconds	Passed	153
100 concurrent users › 100 users park a vehicle at the same moment: every one gets a different slot	Passed	270
100 concurrent users › the slot list shows the 100 new sessions as Occupied	Passed	5
100 concurrent users › 100 concurrent lookups find the right slot among 5,000 vehicles	Passed	110
100 concurrent users › a mix of 100 users (browsing, searching and parking) all succeed together	Passed	192

tests/integration/session-timeout.test.js (12/12 passed)

Test	Result	ms
inactivity timeout › a session idle for 14 minutes still works	Passed	635
inactivity timeout › a session idle for 16 minutes is refused with 401 and deleted	Passed	589
inactivity timeout › a session idle for exactly 15 minutes is refused	Passed	619
inactivity timeout › an expired session is refused on role-protected routes too (401, not 403)	Passed	599
inactivity timeout › an expired session stays refused on the next request	Passed	635

Test	Result	ms
inactivity timeout › the user can log in again after expiry	Passed	749
inactivity timeout › logging out with an expired session still succeeds	Passed	645
activity keeps the session alive › a normal request restarts the idle clock	Passed	671
activity keeps the session alive › a background refresh is allowed but does not extend the session	Passed	628
activity keeps the session alive › a background refresh cannot revive an expired session	Passed	607
other sessions › one expired session does not end another user’s session	Passed	761
other sessions › a made-up cookie is still refused	Passed	532

tests/integration/slot-availability.test.js (27/27 passed)

Test	Result	ms
who can view availability › IT-03_05: a admin can view the slot list	Passed	871
who can view availability › IT-03_05: a driver can view the slot list	Passed	840
who can view availability › IT-03_05: a attendant can view the slot list	Passed	869
who can view availability › without a session the list is refused with 401	Passed	853
who can view availability › the response is never cached	Passed	854
slot statuses › UT-03_01: each slot shows Available, Occupied, Reserved or Unavailable	Passed	880
slot statuses › counts cover every slot	Passed	915
slot statuses › a reservation for later or earlier today does not make a slot Reserved	Passed	844
slot statuses › a cancelled reservation does not make a slot Reserved	Passed	820
slot statuses › Occupied wins over Reserved	Passed	857
slot statuses › Unavailable wins over Occupied	Passed	821
slot statuses › a slot that is reserved right now can still be booked for another time	Passed	830
slot statuses › an occupied slot still cannot be reserved	Passed	830
filtering by status › UT-03_02: ?status=available returns only those slots	Passed	785
filtering by status › UT-03_02: ?status=occupied returns only those slots	Passed	836
filtering by status › UT-03_02: ?status=reserved returns only those slots	Passed	844
filtering by status › UT-03_02: ?status=unavailable returns only those slots	Passed	879
filtering by status › the counts stay the same while filtering	Passed	818
filtering by status › status and type can be combined	Passed	937
filtering by status › an empty filter means all slots	Passed	899
filtering by status › a filter that matches nothing returns an empty list	Passed	902
filtering by status › an unknown status or type is refused with 400	Passed	897
filtering by status › a repeated status parameter is refused with 400	Passed	899
status changes show immediately › IT-03_03: a slot turns Occupied as soon as a vehicle enters	Passed	878
status changes show immediately › IT-03_04: the slot turns Available once the vehicle has exited and paid (UC-08)	Passed	1041
status changes show immediately › a slot turns Unavailable as soon as an admin disables it	Passed	930
NFR-01: the list loads within 2 seconds for 95% of requests › ST-03_06: 300 slots with sessions and reservations load in under 2 seconds at the 95th percentile	Passed	978

tests/integration/slots.test.js (17/17 passed)

Test	Result	ms
adding a slot › a new slot is Available and appears in the list	Passed	666
adding a slot › the list is in natural code order	Passed	631
adding a slot › a code already in use is refused with 409	Passed	568
adding a slot › invalid details are refused with 400 and nothing is saved	Passed	620
disabling a slot › IT-12_05: a disabled slot shows Unavailable and is not assignable	Passed	604
disabling a slot › a disabled slot can be enabled again	Passed	693
disabling a slot › only slots of the requested type are assignable (DR-07)	Passed	655
disabling a slot › disabled must be true or false	Passed	624
disabling a slot › slot id "999" is not found	Passed	654
disabling a slot › slot id "1abc" is not found	Passed	658
only administrators manage slots (NFR-11) › get /api/slots without a session gives 401	Passed	618

Test	Result	ms
only administrators manage slots (NFR-11) › post /api/slots without a session gives 401	Passed	618
only administrators manage slots (NFR-11) › patch /api/slots/1 without a session gives 401	Passed	553
only administrators manage slots (NFR-11) › a driver is refused with 403 and nothing changes	Passed	738
only administrators manage slots (NFR-11) › a attendant is refused with 403 and nothing changes	Passed	767
only administrators manage slots (NFR-11) › a driver may view the slot list	Passed	703
only administrators manage slots (NFR-11) › a attendant may view the slot list	Passed	742

tests/integration/validation.test.js (11/11 passed)

Test	Result	ms
registration input › an empty body is rejected per field and nothing is stored	Passed	542
registration input › a request with no body at all is rejected	Passed	397
registration input › values of the wrong type are rejected	Passed	498
registration input › a password over 72 bytes is rejected	Passed	403
registration input › a name that is too short is rejected	Passed	410
login input › an empty email and password are rejected with field messages	Passed	393
login input › a password over 72 bytes is rejected	Passed	383
request body › malformed JSON gets a clear 400	Passed	411
request body › a body over the size limit gets a clear 413, not a 500	Passed	383
session cookie input › a cookie that parses to an object is treated as not logged in	Passed	381
session cookie input › logout with such a cookie still succeeds	Passed	359

tests/integration/vehicle-entry.test.js (15/15 passed)

Test	Result	ms
recording entry (FR-05) › UT-05_01: entry creates an active session with the entry time recorded	Passed	24
recording entry (FR-05) › UT-05_02: an unregistered vehicle is refused with "Vehicle not found"	Passed	19
recording entry (FR-05) › IT-05_03: a vehicle with an active session cannot enter again (DR-02)	Passed	50
recording entry (FR-05) › IT-05_04: entry is refused when no slot is free	Passed	19
recording entry (FR-05) › IT-05_05: the entry time matches the system clock	Passed	23
recording entry (FR-05) › ST-05_06: entry completes within 3 seconds (NFR-02)	Passed	23
recording entry (FR-05) › UT-06_02: a two-wheeler gets a two-wheeler slot (DR-07)	Passed	34
the assigned slot is consistent everywhere (FR-06) › ST-06_06: after entry the slot shows Occupied for every role, in tracking and in the report	Passed	46
checking the registration number › undefined is refused with "Enter a registration number"	Passed	11
checking the registration number › " " is refused with "Enter a registration number"	Passed	3
checking the registration number › "1234" is refused with "Enter a valid registration number"	Passed	4
checking the registration number › ["KA01AB1234"] is refused with "Enter a valid registration number"	Passed	4
only attendants record entry (NFR-11) › without a session the request is refused with 401	Passed	2
only attendants record entry (NFR-11) › a driver is refused with 403 and no session starts	Passed	18
only attendants record entry (NFR-11) › a admin is refused with 403 and no session starts	Passed	15

tests/integration/vehicle-exit.test.js (20/20 passed)

Test	Result	ms
recording exit (FR-08) › UT-08_01: exit records the time and the session waits for payment	Passed	41
recording exit (FR-08) › UT-08_02: the slot is released to Available once the exit is paid (SYS-12, UC-08)	Passed	51
recording exit (FR-08) › IT-08_03: exit is blocked when there is no active session (DR-04)	Passed	23
recording exit (FR-08) › IT-08_04: an exit time before the entry time is rejected (DR-09)	Passed	23
recording exit (FR-08) › IT-08_05: exit calculates the fee before payment completes the session	Passed	30
recording exit (FR-08) › ST-08_06: a recorded exit cannot be changed by another exit (DR-10)	Passed	44
recording exit (FR-08) › two exits at the same moment record the exit only once	Passed	43
exit input and access › an exit time in the future is rejected	Passed	38
exit input and access › an exit time that is not a date is rejected	Passed	30
exit input and access › an exit time that is an impossible date is rejected	Passed	27

Test	Result	ms
exit input and access › an exit time that is a number is rejected	Passed	30
exit input and access › a missing registration number is rejected	Passed	17
exit input and access › an administrator can record exit	Passed	27
exit input and access › a driver cannot record exit, and a visitor must log in	Passed	29
fee on exit (FR-09) › UT-09_01: 10:00 to 12:00 at ₹20/hr costs ₹40	Passed	33
fee on exit (FR-09) › UT-09_02: 10:00 to 10:45 at ₹20/hr is rounded up to the 30-minute billing step	Passed	40
fee on exit (FR-09) › IT-09_03: 1.5 hours at ₹30/hr costs ₹45	Passed	46
fee on exit (FR-09) › IT-09_04: entry and exit at the same time cost the minimum fee, with no error	Passed	34
fee on exit (FR-09) › IT-09_05: a new exit uses the rate as changed (admin side: IT-12_04)	Passed	61
fee on exit (FR-09) › ST-09_06: the fee is shown within the 3-second exit flow (NFR-02)	Passed	30

tests/integration/vehicles.test.js (16/16 passed)

Test	Result	ms
registering vehicles › UT-02_01: a new vehicle is saved and listed	Passed	14
registering vehicles › UT-02_02: the driver sees only their own vehicles	Passed	21
registering vehicles › IT-02_03: the same number is refused, even written with spaces and hyphens	Passed	21
registering vehicles › IT-02_04: an invalid registration number is refused	Passed	7
registering vehicles › the number is stored normalised, so entry and locate can find it	Passed	7
registering vehicles › a type that no slot has, such as "Car", is refused	Passed	7
changing vehicles › IT-02_05: changing the colour from Red to Blue is saved	Passed	13
changing vehicles › ST-02_06: the change is still there after logging out and in again	Passed	75
changing vehicles › a vehicle can be deleted	Passed	19
changing vehicles › another driver cannot read, change or delete someone else's vehicle	Passed	18
changing vehicles › vehicle id "1abc" is not found	Passed	12
changing vehicles › vehicle id "0" is not found	Passed	11
changing vehicles › vehicle id "999999" is not found	Passed	12
only drivers use My vehicles (NFR-11) › without a session the request is refused with 401	Passed	6
only drivers use My vehicles (NFR-11) › a attendant is refused with 403	Passed	120
only drivers use My vehicles (NFR-11) › a admin is refused with 403	Passed	119